

$$\textcircled{T} \quad \exists f \text{ на } [a, b] \Rightarrow \underline{I = \int_a^b f(x) dx} \quad (\Leftrightarrow)$$

$$\forall \{(\tau^n, \xi^n)\} : \lambda(\tau^n) \xrightarrow{n \rightarrow \infty} 0 \quad \text{бон-се}$$

$$\boxed{\mathcal{S}_{\tau^n}(f, \xi^n) \xrightarrow{n \rightarrow \infty} I}$$

$$\Delta \textcircled{H} \quad \exists \quad I \stackrel{\text{def}}{=} \int_a^b f(x) dx \xRightarrow{\text{def}} \forall \varepsilon > 0 \quad \exists \delta = \delta(\varepsilon) > 0:$$

$$\forall (\tau, \xi) : \lambda(\tau) < \delta \Rightarrow |\mathcal{S}_{\tau}(f, \xi) - I| < \varepsilon$$

$$\mathcal{D} \{(\tau^n, \xi^n)\} : \lambda(\tau^n) \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \exists \delta > 0 \quad \exists N_{\delta}(\delta) \in \mathbb{N}$$

$$\forall n > N_{\delta} \Rightarrow \lambda(\tau^n) < \delta \Rightarrow \text{тпу } n > N_{\delta} \quad \text{бон}$$

$$\underline{|\mathcal{S}_{\tau^n}(f, \xi^n) - I| < \varepsilon \xRightarrow{\text{def}} \lim_{n \rightarrow \infty} \mathcal{S}_{\tau^n}(f, \xi^n) = I}$$

$$\textcircled{g} \quad \exists \forall \{(\tau^n, \xi^n)\} : \lambda(\tau^n) \xrightarrow{n \rightarrow \infty} 0 \quad \textcircled{**} \quad \text{бон-се}$$

$$\mathcal{S}_{\tau^n}(f, \xi^n) \xrightarrow{n \rightarrow \infty} I \quad \textcircled{***} \quad \text{но применим противоположн.$$

$$\text{закл-е: } I \neq \int_a^b f(x) dx \quad (\Rightarrow) \quad \exists \tilde{\varepsilon} > 0 : \forall \delta > 0$$

$$\exists (\tau^{\delta}, \xi^{\delta}) : \lambda(\tau^{\delta}) < \delta \quad \text{и} \quad |\mathcal{S}_{\tau^{\delta}}(f, \xi^{\delta}) - I| > \tilde{\varepsilon}$$

$$\exists \delta_n = \frac{1}{n} \Rightarrow \exists (\tau^n, \xi^n) : \lambda(\tau^n) < \frac{1}{n} \quad \text{и}$$

$$|\mathcal{S}_{\tau^n}(f, \xi^n) - I| > \tilde{\varepsilon} \quad \textcircled{*}$$

$$\text{Т.к. } \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \Rightarrow \lim_{n \rightarrow \infty} \lambda(\tau^n) = 0 \Rightarrow$$

$$\{(\tau^n, \xi^n)\} - \text{фф-т укл.} \quad \textcircled{**}, \text{ но в то же время}$$

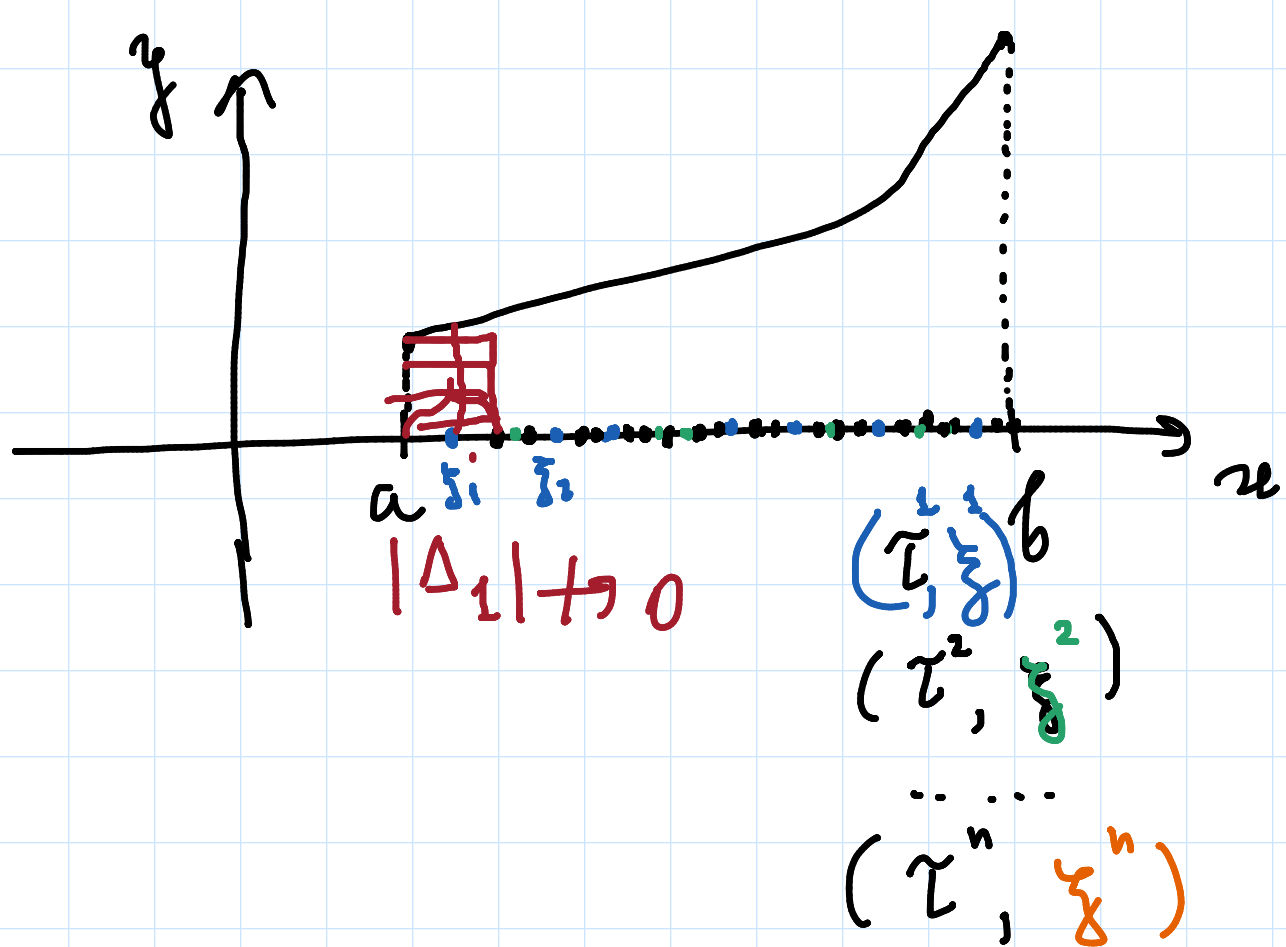
$$\text{бон-се} \quad \textcircled{*} \Rightarrow \mathcal{S}_{\tau^n}(f, \xi^n) \not\xrightarrow{n \rightarrow \infty} I - \text{это пр-ме}$$

с $\textcircled{***}$ Δ

Опр Ф-ция $f(x)$ наз. интегрируемой по Риману на $[a, b]$, если

$$\exists \int_a^b f(x) dx \quad (\text{инт. Римана})$$

Опр Класс инт-х по Р. Ф-ций на $[a, b]$ об-се $R[a, b]$ ($f(x) \in R[a, b]$)



$n \rightarrow \infty$

$$S_n(f, \xi) \xrightarrow{n \rightarrow \infty} I ?$$

$$\lambda(\tau^n) \rightarrow 0$$

Tip. 1) $f(x) = C \in \mathbb{R} [a, b]$ u

$$\int_a^b C dx = C(b-a) ?$$

$$\Delta \quad S_n(f, \xi) = \sum_{i=1}^n C \Delta x_i = C \sum_{i=1}^n \Delta x_i = C(b-a)$$

$$\Rightarrow \lim_{\substack{n \rightarrow \infty \\ (\lambda \rightarrow 0)}} S_n(f, \xi) = C(b-a)$$

$$2) f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{I} \end{cases} \quad] x \in [0, 1]$$

$$\begin{aligned}] \xi_i \in \mathbb{Q} & \Rightarrow S_n(f, \xi) = \sum_{i=1}^n 1 \cdot \Delta x_i = 1 \\] \xi_i \in \mathbb{I} & \Rightarrow S_n(f, \xi) = \sum_{i=1}^n 0 \cdot \Delta x_i = 0 \end{aligned}$$

$$\Rightarrow \begin{cases} \lim_{\substack{\lambda(\xi) \rightarrow 0 \\ (n \rightarrow \infty)}} S_n(f, \xi) = 0 \\ \lim_{\substack{\lambda(\xi) \rightarrow 0 \\ (n \rightarrow \infty)}} S_n(f, \xi) = 1 \end{cases}$$

$$\Rightarrow f(x) \notin R[a, b]$$

Onp

$$\int_a^a f(x) dx = 0$$

$$\int_b^a f(x) dx = - \int_a^b f(x) dx, \quad a < b$$

Опр f на $[a, b]$, τ - разб. $[a, b]$

$$S_{\tau}(f) = \sum_{i=1}^n M_i \Delta x_i - \text{верхняя}$$

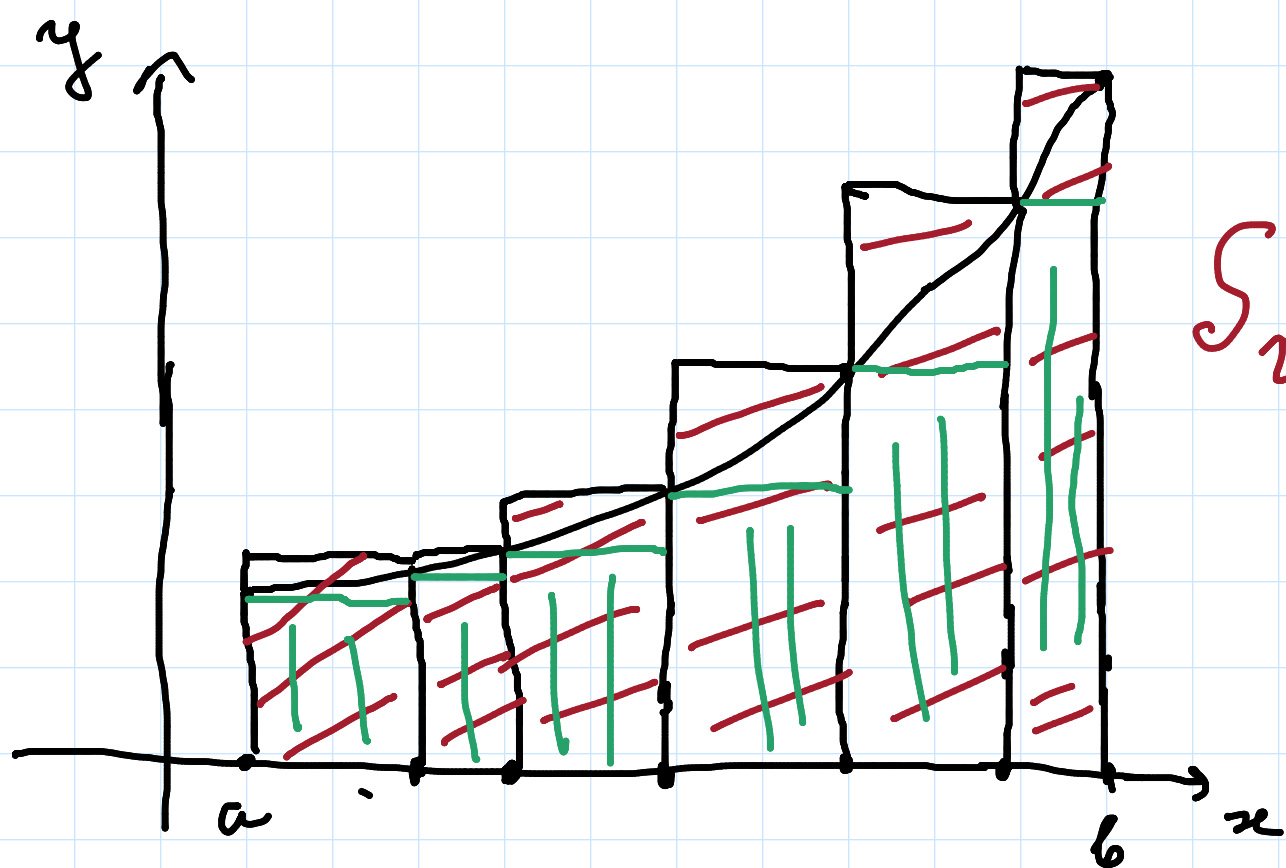
$$M_i = \sup_{\Delta_i} f(x)$$

сумма Дарбу
при разб-м τ

$$S_{\tau}(f) = \sum_{i=1}^n m_i \Delta x_i - \text{нижняя}$$

$$m_i = \inf_{\Delta_i} f(x)$$

сумма Дарбу
при разб-м τ



$$S_{\tau}(f)$$

$$s_{\tau}(f)$$

Сб-бо (1)

$$s_{\tau}(f) \leq \mathcal{D}_{\tau}(f, \xi) \leq S_{\tau}(f)$$

$$\Delta \quad \sum_{i=1}^n m_i \Delta x_i \leq \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \sum_{i=1}^n M_i \Delta x_i \quad ?$$

$$\text{Т.к. } \forall \xi_i \in \Delta_i \Rightarrow m_i \leq f(\xi_i) \leq M_i$$

$$\Rightarrow m_i \Delta x_i \leq f(\xi_i) \Delta x_i \leq M_i \Delta x_i, \quad i = \overline{1, n}$$

$$\sum_{i=1}^n m_i \Delta x_i \leq \sum_{i=1}^n f(\xi_i) \Delta x_i \leq \sum_{i=1}^n M_i \Delta x_i$$

Сб-бо (2)

$$S_{\tau}(f) = \sup_{\xi} \mathcal{D}_{\tau}(f, \xi)$$

$$s_{\tau}(f) = \inf_{\xi} \mathcal{D}_{\tau}(f, \xi)$$

$$\Delta \quad \text{Ф-н: } S_{\tau}(f) = \sup_{\xi} \mathcal{D}_{\tau}(f, \xi) \quad ?$$

$$1) \textcircled{1} \text{ } f \text{ - op. ch } \Rightarrow \forall \xi \stackrel{\textcircled{1}}{=} \partial_{\tau}(f, \xi) \leq S_{\tau}(f) \text{ (bun)}$$

$$\forall \varepsilon > 0 \exists \xi' : \partial_{\tau}(f, \xi') > S_{\tau}(f) - \varepsilon ?$$

$$\text{T.K. } M_i = \sup_{x \in \Delta_i} f(x) \Rightarrow \forall \varepsilon > 0 \exists \xi'_i = \xi'_i(\varepsilon) \in \Delta_i :$$

$$\Rightarrow M_i - \frac{\varepsilon}{b-a} < f(\xi'_i), \quad i = \overline{1, n}$$

$$\sum_{i=1}^n M_i \Delta x_i - \sum_{i=1}^n \frac{\varepsilon}{b-a} \Delta x_i < \sum_{i=1}^n f(\xi'_i) \Delta x_i$$

$$\Rightarrow S_{\tau}(f) - \varepsilon < \partial_{\tau}(f, \xi') \Rightarrow$$

$$S_{\tau}(f) = \sup_{\xi} \partial_{\tau}(f, \xi)$$